

Taylor Wang

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Portfolio: <https://taylwang.github.io/portfolio/>

EDUCATION

Columbia University

Degree: B.S. in Electrical Engineering

GPA: 3.87/4.00

New York, NY

Expected May 2027

RESEARCH EXPERIENCE

NASA Jet Propulsion Laboratory

Role: Microwave Systems Intern

Mentors: Dr. Omkar Pradhan and Dr. Maryam Salim

Jun 2025–Aug 2025

& Jun 2026–Aug 2026

La Cañada, CA

- Identified a misaligned via backdrill on a polarimetric spectrometer PCB as the cause of unexpected attenuation in its frequency response.
- Formulated a mathematical model of the spectrometer's input signals incorporating ADC gain and offset mismatches to explain periodic spurious peaks in the frequency spectrum.
- Developed an FFT-based metric to quantify the strength of impedance-mismatch-induced ripples in the spectrometer's frequency response and used it to assess attenuator effectiveness for ripple suppression.
- Characterized the spectrometer's front-end amplifier by calculating its noise figure and gain using Y-factor measurements and confirming the gain with S-parameter measurements.
- Diagnosed a balun-induced signal inversion in the radiometer responsible for correlated signals appearing as anti-correlated.

Columbia Wireless & Mobile Networking Lab

Role: Undergraduate Researcher

Advisor: Prof. Gil Zussman

Oct 2025–Jan 2026

New York, NY

- Developed and trained a neural network to replicate an existing model's conversion of radiometer brightness temperatures into atmospheric water vapor profiles, enabling modification of the model to study its robustness to radio-frequency interference.
- Classified certain K-band frequency channels as unreliable after identifying missed liquid-nitrogen calibrations as the likely cause of incomplete data coverage in those channels.
- Developed a Python pipeline to detect 5G radio-frequency interference across radiometer channels by identifying unusually high brightness temperature measurements.

Columbia Solar Fuels Lab

Role: Undergraduate Researcher

Advisor: Prof. Daniel Esposito

Sep 2024–May 2025

New York, NY

- Employed cyclic voltammetry and chronoamperometry to analyze the reactivity of a hydrogen peroxide-producing flow cell using different catalysts, electrolytes, and separators.
- Determined that oxygen-limited reaction kinetics constrained hydrogen peroxide production, redirecting subsequent experiments toward improving oxygen mass transfer rather than electrolyte composition or charge transfer.
- Recognized that oxygen needed for the cathodic reaction was pushed in the direction of electrolyte flow, prompting the placement of the cathode near the output.

UW-Madison Woods Group

Role: Computational Assistant

Advisor: Prof. Benjamin Woods

May 2024–Sep 2024

Madison, WI

- Solved the 2D Schrödinger equation numerically using finite-difference methods and sparse matrix techniques to obtain quantum-state energies and wave functions.
- Separated contributions from distinct quantum valleys by constructing a localized basis, enabling the calculation of intra-valley and inter-valley coupling.

Mad City Labs

Role: Programmer

Supervisor: Jim MacKay

May 2023–Aug 2023

Madison, WI

- Developed a Java-based graphical user interface for controlling company microscope systems.
- Integrated open-source microscope control modules into the user interface, eliminating the need to develop communication protocols for four hardware devices.

RESEARCH PROJECTS

Search Algorithm for the Lost Mars Polar Lander

Jan 2026–Present

- Developing a computational imaging pipeline to search Mars orbital imagery for NASA's lost Mars Polar Lander through image registration, super-resolution, and object identification.
- Implemented Adaptive Least-Squares Correlation to refine corresponding tie points between Mars orbital images by iteratively estimating local affine transformation parameters.
- Implemented a Bayesian multi-frame super-resolution algorithm using MAP estimation and a Field-of-Experts image prior to reconstruct high-resolution Mars orbital imagery.

Automated Left Ventricle Identification in Echocardiograms

Sep 2025–Dec 2025

- Developed an automated image-processing pipeline in Python to identify the left ventricle in echocardiograms using image enhancement, watershed segmentation, and region-based classification, achieving 85% accuracy.
- Designed median and anisotropic Gaussian filtering methods to enhance ventricle boundaries while suppressing imaging artifacts.
- Devised a weighted scoring algorithm incorporating centroid location, area, and shape to identify the left ventricle among segmented candidate regions.

Nickel Hydroxide Catalyst for Ammonia Oxidation

Jan 2022–Oct 2022

- Electrochemically deposited nickel hydroxide onto nickel substrates for ammonia oxidation in a home laboratory.
- Evaluated the effects of cathode current density and $\text{Ni}(\text{NO}_3)_2$ concentration on nickel hydroxide deposition, identifying 0.085 M $\text{Ni}(\text{NO}_3)_2$ and 1.1 mA/cm² as effective conditions.
- Designed a substitute current source for the electrochemical cell by regulating the current from a 9V battery through a voltage-divider circuit, with an Arduino adjusting the voltage division based on measured current output.

TEACHING EXPERIENCE

Teaching Assistant

Course: Intro to Electrical Engineering, Columbia University

Sep 2025–Present

New York, NY

- Guided undergraduate students through laboratory experiments in an electrical engineering course and assisted with experimental setup, measurement, and analysis.
- Evaluated reports and provided feedback on experimental methods and conceptual understanding.

HONORS AND AWARDS

The Gates Scholarship, The Bill and Melinda Gates Foundation	2023
Soemitro Scholarship Fund, Columbia University	2026
Tau Beta Pi Engineering Honor Society Candidate, Tau Beta Pi Association	2025
\$1,500 Exceptional Merit Scholarship, Oxbridge Academic Programs	2022
\$5,280 Global Navigator Scholarship, Council on International Educational Exchange	2021

SKILLS

Core Competencies: RF/Microwave Engineering, Wireless Communications, Digital Signal Processing, Image Processing, Computer Vision, Machine Learning

Programming: Python, Java, MATLAB

Laboratory Equipment: Vector Network Analyzer, Spectrum Analyzer, Oscilloscope, Signal Generator

RELEVANT COURSEWORK

Communication Circuits; Digital Communications; Digital Signal Processing; Mathematics of Machine Learning, Signals, and Control; Machine Learning for Signals, Information, and Data.